

Ayush Kumar

Senior Software Engineer | Oracle Cloud Infrastructure | Kubernetes | Distributed Systems

Hyderabad, India | +91 8168063477 | ayushkumardudi1999@gmail.com | LinkedIn: linkedin.com/in/ayush-kumar-73481a179

SUMMARY

- Senior Software Engineer at Oracle with 4+ years building OCI management-plane services, Kubernetes and OKE lifecycle automation, Terraform provisioning, and workflow orchestration systems.
- Specializes in distributed workflow systems, infrastructure automation, rollout safety, production observability, and operational hardening for large-scale internal platforms.
- Pragmatic production engineer with experience taking platform features from design and implementation through release, support, and iterative reliability improvements.

TECHNICAL SKILLS

Languages: Go, Bash, Python

Cloud and Infrastructure: OCI, OKE, Kubernetes, Helm, CRD-backed integrations, client-go, controller-runtime client, Terraform/OpenTofu, OCI DevOps/Build, Shepherd

Distributed Systems: Temporal workflows, reconciliation loops, CI/CD, release automation, service orchestration, Cluster Autoscaler concepts

Observability and Operations: Mimir/Prometheus alerting, runbooks, production debugging, metrics, release validation

Datastores and Tooling: MySQL, Git, Bitbucket, OCI SCM, Docker

PROFESSIONAL EXPERIENCE

Oracle | Senior Software Engineer | Hyderabad, India | August 2024 - Present

- Designed and developed an OCI DevOps custom-image pipeline for node-management Kubernetes nodes, reducing node creation time from about 20 minutes to about 5 minutes through prebaked images.
- Led OKE-to-node-management migration enablement across service, configuration, platform, and runbook layers for about 111K production nodes.
- Built Go-based Temporal workflows for OKE and node-management lifecycle reconciliation, covering node pool discovery, image selection, work-request polling, scale/rotate, node replacement, cordon/drain/delete, and retryable state progression.
- Owned node-management observability and reliability signals across 38 production regions, covering production alerting, dashboards, and request health over a 7-day production window.
- Served in the platform on-call rotation, triaging incidents and resolving user issues for an internal platform supporting 2,237 users.

Oracle | Applications Engineer | Hyderabad, India | July 2021 - August 2024

- Built Terraform Worker and provisioning workflows for OCI platform automation, covering init/plan/apply execution, backend state, provider management, workload identity authentication, CLI support, and production policies.
- Bootstrapped node-management production readiness through platform integration, Shepherd configuration, configurable retries and timeouts, release automation, fleet whitelisting, and operational runbooks.
- Supported high-volume workflow execution paths handling about 48M node-management Temporal activities and about 153K Terraform Temporal activities over a 7-day production window.
- Deployed third-party CRD-backed Kubernetes components through Helm, including Gatekeeper, cert-manager, Traefik, ServiceMonitor integrations, middleware, and DP platform releases.
- Built workflow-driven MySQL backup/restore and Terraform automation, including migration support from etcd to MySQL.
- Improved delivery quality through performance refactoring, release validation, production debugging, operational tooling, and rollout support.

INTERNSHIPS

Next Education India Pvt Ltd | Technology Intern | Hyderabad, India | May 2020 - July 2020

- Developed a machine comprehension web service using AllenNLP and Django for automatic question answering.
- Improved reading comprehension models with ELMo embeddings, BiDAF, and NAQANet, and containerized the service with Docker.

Indian Institute of Science (IISc) | Research Intern | Bengaluru, India | May 2018 - July 2018

- Worked on improving the efficiency of oscillating airfoils for autonomous underwater vehicle applications.
- Studied propulsion techniques, chordwise flexibility of NACA-0012 airfoils, non-sinusoidal gaits, and wake velocity fields using PIV with Nd-YAG laser.

EDUCATION

Indian Institute of Technology, Roorkee | B.Tech, Mechanical Engineering | June 2017 to June 2021